

```

# Pick the mom haplotype that was transmitted to the kid.
df_meth = df_meth.with_columns(
    mom_transmitted = pl.when(pl.col("mat_hap") == "C")
        .then(pl.col("mom_C"))
        .otherwise(pl.col("mom_D"))
)

# Find runs of consecutive CpGs where the kid_mat is unmethylated
# but the mom-transmitted haplotype is methylated - i.e., a
# *loss* of methylation confined to the maternally transmitted
# homolog (the canonical case at maternally methylated imprinting
# control regions).
WINDOW = 2
df_lom_mat = (
    df_meth.with_columns(
        delta = pl.col("kid_mat") - pl.col("mom_transmitted"),
    )
    .with_columns(
        mean_delta = pl.col("delta").rolling_mean(WINDOW),
    )
    .filter(pl.col("mean_delta") < -0.5)
)

```